generator that combines low power consumption and cost-efficiency in a small package. The VersaClock 7 can simplify the system design by replacing multiple discrete timing components and lower costs by reducing bill of material and board space. It enables users to configure frequencies, I/O levels, and GPIO pins. The programmable clock comes in a compact space-saving package. Furthermore, the high levels of integration and the flexibility of programmability help manufacturers reduce component counts, saving board space and power. These features offer greater versatility in consumer, data communications, telecommunications, and networking applications.

Renesas
https://www.renesas.com

Battery charging IC

Infineon's ICC80QSG is a single-stage flyback controller for the safe and robust AC-DC battery charger. The IC offers a quasi-resonant mode switching in valley n (QRMn). Its QRM operation comes with continuous conduction mode (CCM) prevention and valley switching discontinuous conduction mode (DCM) in mid-to-light load. It features a reduced gate driver output voltage during burst mode, making it ideally suited for very low standby requirements under light load or no-load conditions with a remarkable standby performance over the whole operating range. The charging IC offers a comprehensive set of protection features, which enable engineers to design safe and robust AC-DC battery chargers for a variety of applications.

Infineon
https://www.infineon.com

BOARDS AND MODULES

High-power bus converter

Artesyn NDQ900 is a non-isolated bus converter (NIBC). The digital DC/DC converter comes in an industry-standard quarter-brick format for 48V power conversion in compute and telecom applications. The bus converter can supply power of up to 900W with a high efficiency of up to 97% across a wide load range. The NDQ900 NIBC features a PMBus interface for flexible digital control and monitoring. The module also offers a remote-control function and a good power signal. The Artesyn NDQ900 is suitable for use in test and measurement equipment, data storage, telecommunication, high computing equipment, etc.

Artesyn
https://www.artesyn.com

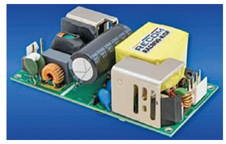
Gain blocks for radar

The CMX90G701 and CMX90G702 are positive gain-slope amplifiers that are suited to a range of radio frequency (RF) and wireless applications. Both the amplifiers are suitable for wireless applications operating in the 6 to 18GHz frequency range. The amplifiers are capable of compensating the system losses, thus eliminating the need for passive equalisers. Both devices deliver a small signal gain within the range of 9.5 to 11.5dB, a P1dB output rating of +10.0dBm, and a low noise figure of 3dB. These devices are fabricated using GaAs pHEMT process technology to achieve an optimal combination of high linearity, low noise, and low DC power consumption. The CMX90G701 and CMX90G702 are suitable for use in applications such as satellite communications, 5G fixed wireless access (FWA), microwave backhaul, radar, etc.

CML Microcircuits
www.cmlmicro.com

90W AC/DC power supply

RECOM's new 90W RACM90-K AC/DC power supply is a versatile power supply suitable for cost-sensitive applications. The 90W power supply can operate over a wide range of input voltages ranging from 85-264V AC and provides a tightly regulated single output of 12V, 15V, 24V, 36V, and 48V DC. The module is available in the industry-standard 5cm × 10cm format. The power supply features dual fuses on-board and offers immunity to surges for installation Class 3 and over-voltage category OVCIII. The module has been rated to operate at an altitude of up to 4,000 metres and can work over a wide range of operating temperatures from -40°C to +90°C. The power supply is suitable for use in demanding applications, such as medical, industrial, and household applications.



Recom Power
www.recom-power.com

Voltage regulator for CPUs

The new BMR510 is a 2-phase integrated power stage which delivers a continuous total output current of 40A per phase or 80A total with an efficiency of up to 90%. Compared to the previous model the upgraded module offers higher inductance in a smaller package, thus allowing a lower switching frequency operation. The module allows for remote monitoring and comes with protective features, including output over-current and over-temperature protection. The BMR510 is optimised for top-side cooling and is halogen-free. It can accept tri-state pulse width modulation inputs from the chosen controller and a separate enable input is provided. The device